

V.S.B. ENGINEERING COLLEGE, KARUR

(An Autonomous Institution)

DEPARTMENT OF INFORMATION TECHNOLOGY

PROJECT REPORT

ON

**EMPLOYEE PAYROLL & WORKFORCE MANAGEMENT
SYSTEM**

Enterprise Employee Directory, Automated Leave Tracker & Payroll Processor

Submitted by,

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III B.Tech Information Technology

**BACHELOR OF TECHNOLOGY
IN
INFORMATION TECHNOLOGY**

Academic Batch: 2024 - 2028

BONAFIDE CERTIFICATE

This is to certify that the project report entitled "**Employee Payroll & Workforce Management System**" is a bonafide record of work done by **Kavi Bharathi P** (Register Number: **922524205074**) of the V Semester, Department of Information Technology, in partial fulfilment of the requirements for the project work during the academic year 2025–2026, at V.S.B. Engineering College, Karur (An Autonomous Institution).

The report has been verified and found satisfactory. It is submitted for evaluation as part of the curriculum of the Department of Information Technology.

DECLARATION

I, **Kavi Bharathi P** (Register Number: **922524205074**), student of V Semester, Department of Information Technology, V.S.B. Engineering College, Karur, hereby declare that the project report entitled "**Employee Payroll & Workforce Management System**" submitted in partial fulfillment of B.Tech curriculum requirements is a record of original project work carried out by me under the guidance of my project supervisor. This work has not previously formed the basis for the award of any other degree, diploma, associate-ship, or fellowship in this or any other university.

Place: Karur

Date:

Signature of the Student

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Kavi Bharathi P

ABSTRACT

Modern workforce administration demands automated tools to streamline administrative overhead, secure sensitive employee data, and process financial benefits. Traditional paper-based tracking or legacy spreadsheets are prone to data corruption, human errors, lack of transparency, and high security vulnerabilities. To address these limitations, we present the *Employee Payroll & Workforce Management System*, a full-stack, enterprise-grade, responsive platform designed to consolidate employee directory services, automated attendance tracking, dynamic leave requests, and robust payroll processing.

The platform is architected using a modern decoupled three-tier software architecture. The frontend is built on **React 19** and **Vite 8.0**, employing a clean component design styled with **Tailwind CSS 4.3**. This responsive layout features a slate-dark theme with micro-animations for optimized user experiences across desktop, tablet, and mobile browsers. The backend server is powered by **Spring Boot 3.3.5** and **Java 21**, implementing stateless JWT-based token authentication via **Spring Security** and object-relational mapping through **Spring Data JPA**. Data storage is handled via a relational **MySQL 8.0** engine, structured with comprehensive foreign key constraints and indexes for transactional integrity.

Key capabilities include automated email OTP verification for secure login credentials, automatic check-in/check-out logs, intelligent leave balances tracking, payroll computation with dynamic allowances and deductions, and automatic client-side downloadable PDF payslips compiled using a cryptographic validation layout. The platform implements rigorous role-based access control (RBAC) separating Admin, HR, and Employee actions, ensuring tamper-proof operations. The resulting codebase is modular, production-tested, and serves as a highly scalable solution suitable for corporate and academic deployments.

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1. INTRODUCTION

Efficient administrative coordination and financial compliance are crucial pillars for corporate productivity and structural stability. In modern corporations, tracking employee performance, attendance patterns, leaves, and financial payroll computation presents a complex administrative challenge. Historically, corporate entities operated on manual paper files, check-in registries, and legacy spreadsheets to compute basic salaries and generate paper payslips. However, manual workflows suffer from severe administrative overhead, susceptibility to calculations anomalies, lack of real-time auditing, and massive security vulnerabilities regarding confidential information.

As organizations scale, the need for automated, secure, and unified software architectures becomes imperative. The integration of real-time attendance tracking with leave requests and payroll calculators ensures mathematical precision, minimizes disputes, and reduces processing time from days to a few clicks. Furthermore, implementing robust encryption, role-based controls, and automated OTP verifications protects sensitive personal details from malicious data leaks.

The project, **Employee Payroll & Workforce Management System**, represents a state-of-the-art Web Application designed to solve these problems. By bridging responsive React client interfaces with secure Spring Boot servers, the application provides a centralized portal where employees, HR managers, and administrators coordinate seamlessly. Employees record attendance, apply for leaves, and download payslips; HR managers onboard personnel, review leave requests, and run payroll processes; administrators regulate global system settings and security logs.

This project report details the software engineering lifecycle, database models, API documentation, UI components, security layers, and deployment configurations. The resulting platform offers a scalable, secure, and production-ready solution ready to be integrated into enterprise workflows.

2. PROJECT OVERVIEW

The Employee Payroll & Workforce Management System is organized as a unified hub for workforce directory administration. The backend server is built using Java/Spring Boot and serves secure API endpoints, which are consumed by the React-based Vite client application. Data persistence is managed through a relational MySQL schema. Role-Based Access Control (RBAC) separates actions into Admin, HR, and Employee contexts.

Completed actions include OTP-verified email authentication, automated leave balance subtraction, attendance logs with check-in/check-out tracking, allowance/deduction calculations, automated payslip creation, and interactive profile updates. This decoupling ensures high concurrency, clean API segregation, and secure, tamper-proof operations.

3. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Legacy workforce administration workflows depend on manual inputs, leading to data inconsistencies, time theft (buddy punching), delayed payroll calculations, and lack of secure role boundaries. Confined databases and unencrypted files compromise employee profiles and banking details, making compliance difficult.

Objectives:

- Develop a secure, role-based full-stack web application supporting JWT authentication and OTP validations.
- Deploy automated attendance monitoring with check-in and check-out logs.
- Design a dynamic leave management module with automated balance checks and multi-level approval workflows.
- Implement a payroll calculation engine that handles basic salary, tax deductions, bonuses, and allowance calculations.
- Construct a secure PDF payslip compiler that generates downloadable cryptographic documents for employees.
- Establish clean admin controls to monitor system configurations, user status updates, and audit trails.

4. SCOPE & SYSTEM COMPARISON

Scope: The scope of this project encompasses the design, build, and verification of a web portal for employee management. It includes a secure React frontend using custom Tailwind styling, a Spring Boot server, and MySQL database configuration. The system boundaries cover JWT security filters, OTP distribution, profile updates, check-in tracking, leave calculations, and pdf processing. It excludes external bank transaction integrations, as all financial operations are computed and stored virtually.

4.1 Existing System vs. Proposed System

A detailed comparison of the existing manual workflow versus the proposed automated system is shown below:

Feature / Parameter	Existing Manual System	Proposed Automated System
Authentication	Paper registry/Passwords on sheets	Stateless JWT + Secure Email OTP
Attendance Tracking	Manual registry check-ins	Database logs with automatic timestamps
Leave Requests	Paper forms, manual balance tracking	Online requests, auto balance subtraction
Payroll Computation	Excel calculations (error-prone)	Automated system formula calculations
Payslip Distribution	Printed paper slips (delayed)	Instant downloadable PDF from dashboard
Data Security	Unsecured files, physical damage risks	BCrypt password hashing & SQL indexes
Auditing & Reports	Manual reports compiling (weeks)	Real-time admin/HR analytical dashboards

5. TECHNOLOGY STACK & SYSTEM REQUIREMENTS

The platform utilizes a modern decoupled three-tier technology stack engineered to support high concurrent user requests, scalability, and security compliance. A detailed layer-wise description is listed in the table below:

Layer	Technology & Purpose
Frontend Framework	React 19.2 (JavaScript SPA component library)
Build Tool (Frontend)	Vite 8.0 (Fast bundling & hot module replacement)
Styling System	Tailwind CSS 4.3 (Utility-first styling system)
Client State & HTTP	React Context API (Auth Context), Axios (HTTP client)
Backend Web Framework	Spring Boot 3.3.5 (Java 21 Enterprise Framework)
Security & Auth	Spring Security Core (Stateless JWT auth filters, BCrypt hashing)
Persistence & ORM	Spring Data JPA / Hibernate (Entity mappings & repositories)
Database Engine	MySQL 8.0 Relational Engine
PDF Payslip Engine	OpenPDF 2.0.3 (Java-based cryptographic PDF generation)
Mailing System	Spring Boot Starter Mail (SMTP integration for OTP)

5.1 Hardware & Software Requirements

Component	Software Requirements	Hardware Requirements
Operating System	Windows 10/11, Ubuntu 20.04+, macOS	Multi-core 64-bit CPU
Java Runtime	Java Development Kit (JDK) 21 LTS	8 GB RAM (minimum), 16 GB (recommended)
Node.js Runtime	Node.js v20.0+ & npm package manager	10 GB of free hard disk space
Database	MySQL Server 8.0.40 Relational Database	Network Adapter (for API transactions)
Web Browser	Chrome 120+, Edge 120+, Safari, Firefox	Standard desktop/laptop/mobile device

6. SYSTEM REQUIREMENTS SPECIFICATION

System requirements are separated into Functional Requirements (FR) detailing specific capabilities and Non-Functional Requirements (NFR) defining architectural, performance, and security constraints.

6.1 Functional Requirements

ID	Module / Requirement	Detailed Description
FR-1	Authentication & Security	Authenticate users via email/password and secure JWT access tokens.
FR-2	Email OTP Verification	Send numeric email verification codes for new accounts and password resets.
FR-3	Employee Directory	Onboard personnel, track job designations, departments, basic salaries, reporting lines.
FR-4	Attendance Tracker	Log daily check-in and check-out times, calculate worked hours, track absences.
FR-5	Leave Operations	Check leave balances, apply for leave request, process approvals/rejections.
FR-6	Payroll Calculator	Compute gross salaries, allowances, provident fund, tax deductions, and net payouts.
FR-7	Payslip PDF compiler	Generate and download cryptographically formatted PDF payslips on client-side.
FR-8	System Configurations	Configure system rules (allowances, penalties, default work hours) dynamically.

6.2 Non-Functional Requirements

Category	Architectural Requirement / Criteria
Performance	REST API endpoints must resolve in <200ms under standard local network workloads.
Security	Passwords must be hashed using BCrypt. JWT validation mandatory on all secured API paths.
Reliability	Database connection pooling must maintain high uptime; transactional rollbacks on errors.
Usability	Vibrant slate-dark responsive theme, conforming to WCAG contrast standards for readability.
Portability	The containerized stack (Docker) must run seamlessly across Windows, Linux, and macOS.

7. SYSTEM DESIGN & ARCHITECTURE

The system's modular separation of concerns isolates user interactions from data queries. Figure 1 outlines the Presentation Layer (React client), Application Layer (Spring Boot backend), and Data Layer (MySQL database).

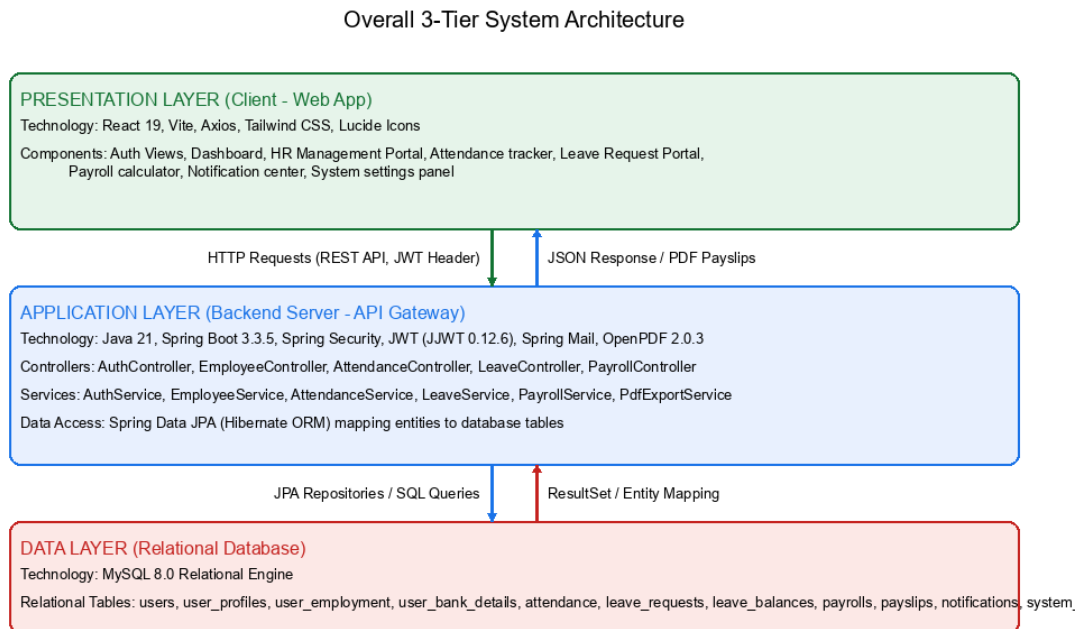


Figure 1: Overall 3-Tier System Architecture Diagram

7.1 System Use Case Boundaries

The functional interactions between actors (Admin, HR Manager, and Employee) and use cases are mapped in Figure 2. Authentication acts as a gateway use case for all actions.

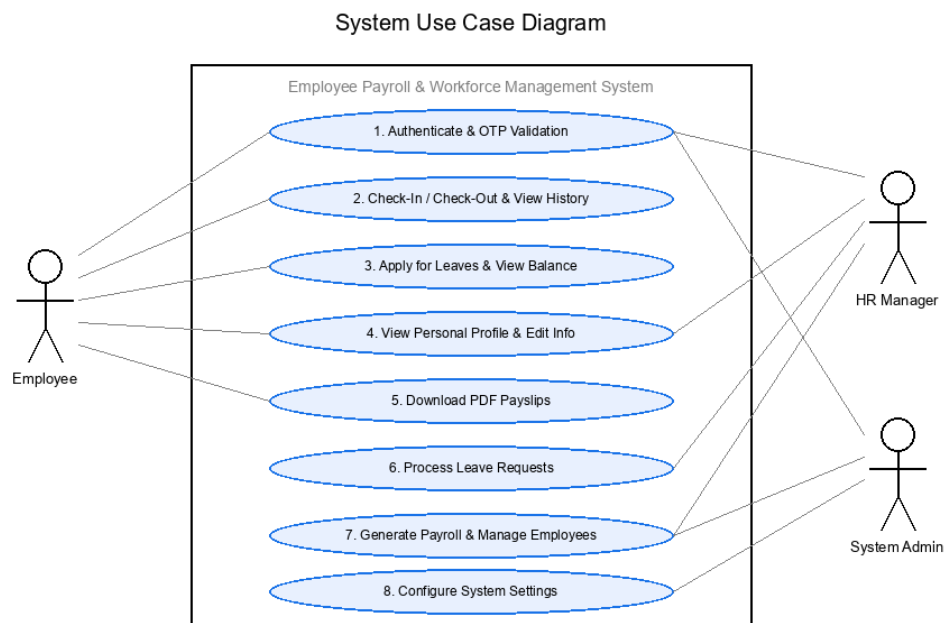


Figure 2: System Use Case Interaction Diagram

7.2 Data Flow Diagram (Level 1)

Figure 3 depicts DFD Level 1, detailing the data exchange between external entities, core processes, and tables. Process 1.0 (Auth) verifies credentials against D1; Process 2.0 (Attendance) updates D2; Process 3.0 (Leaves) reads/writes D3; and Process 4.0 (Payroll) runs computations on D4.

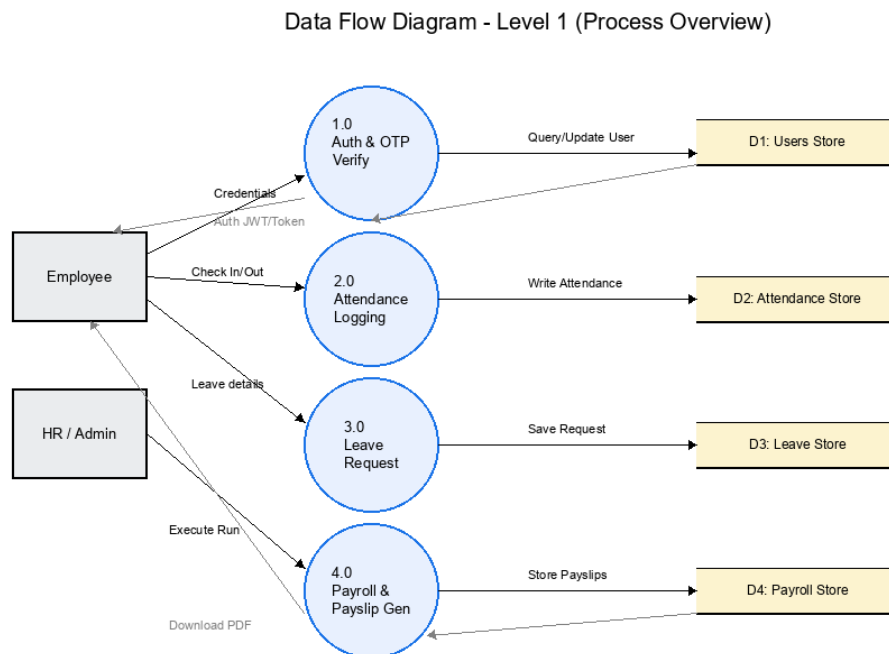


Figure 3: Data Flow Diagram Level 1 Process Model

7.3 Leave Application Workflow

Figure 4 charts the workflow of a leave request, tracing the submission by an employee, validation against leave balances, notification dispatch to HR, approval decision, and balance deductions.

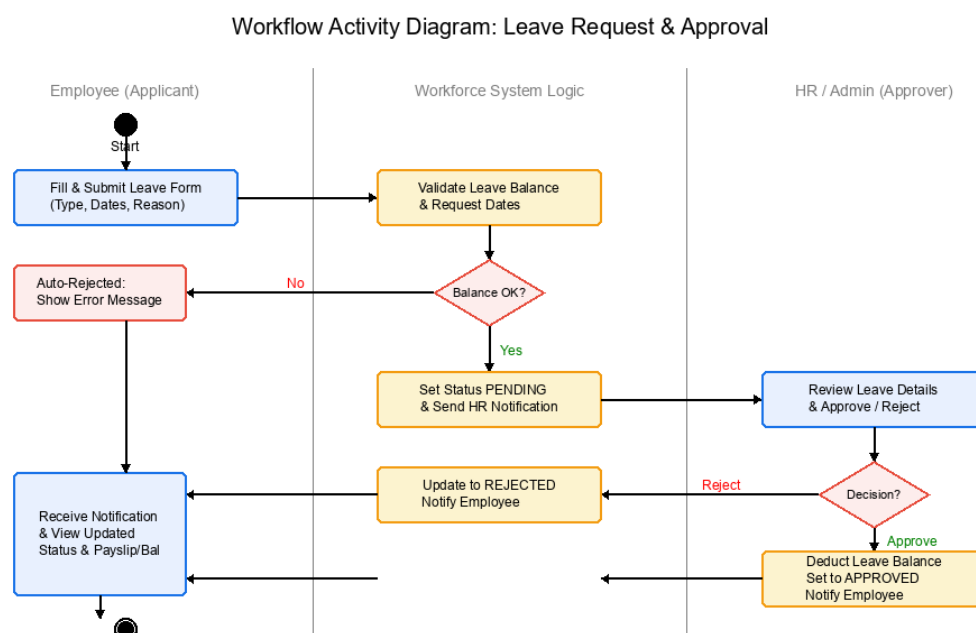


Figure 4: Activity Workflow for Leave Request Processing

8. DATABASE DESIGN & RELATIONAL ERD

The relational database schema is structured to maintain referential integrity. The Users table functions as the root entity, establishing 1-to-1 relationships with user_profiles, user_employment, and user_bank_details, and 1-to-many relationships with attendance, leave_requests, leave_balances, and payrolls. Figure 5 illustrates the relational schema, columns, and foreign keys.

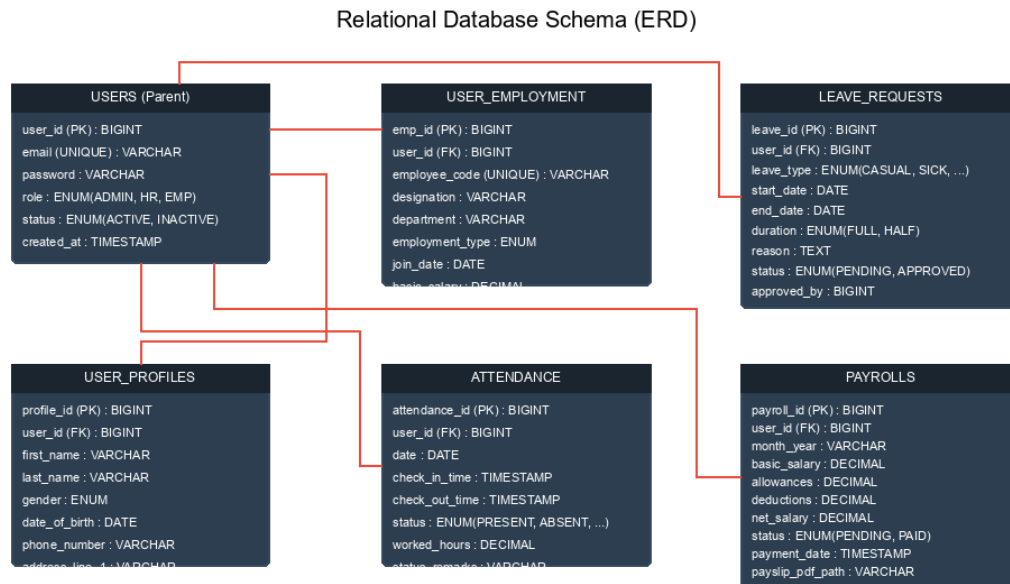


Figure 5: Relational Database Entity-Relationship Diagram (ERD)

8.1 Database Tables & Constraints

Table Name	Primary Key	Foreign Key Mappings	Purpose
users	user_id (BIGINT)	None	Store login email, BCrypt password, status, role.
user_profiles	profile_id (BIGINT)	user_id -> users(user_id)	Personal names, dob, phone, address details.
user_employment	emp_id (BIGINT)	user_id -> users(user_id)	Designation, department, basic salary, join date.
attendance	attendance_id	user_id -> users(user_id)	Daily check-in, check-out timestamps, hours.
leave_requests	leave_id (BIGINT)	user_id -> users(user_id)	Leave type, dates, reason, status, approver.
payrolls	payroll_id (BIGINT)	user_id -> users(user_id)	Gross, deductions, net salary, status, month.

9. SYSTEM REST API ENDPOINTS

The Spring Boot application exposes secure RESTful API endpoints for the React client. All requests except `/api/auth/login`, `/api/auth/signup`, and `/api/auth/reset-password` require a valid Authorization header containing a JWT token. A detailed reference of core endpoints is listed in the table below:

HTTP Method	API Endpoint	Secured?	Functional Description
POST	<code>/api/auth/signup</code>	No	Register new user; dispatches OTP to email address.
POST	<code>/api/auth/verify-otp</code>	No	Verifies signup OTP to activate the user profile.
POST	<code>/api/auth/login</code>	No	Validates password and returns a stateless JWT token.
GET	<code>/api/employees</code>	Yes (HR/Admin)	Fetches list of all employees in database.
POST	<code>/api/employees</code>	Yes (Admin)	Creates and onboards a new employee record.
POST	<code>/api/attendance/check-in</code>	Yes	Logs check-in timestamp with current client date.
POST	<code>/api/attendance/check-out</code>	Yes	Logs check-out timestamp; calculates worked hours.
GET	<code>/api/leaves/balance</code>	Yes	Returns remaining, used, and total days per leave type.
POST	<code>/api/leaves/request</code>	Yes (Employee)	Submit leave request; updates status to PENDING.
PUT	<code>/api/leaves/{id}/approve</code>	Yes (HR/Admin)	Approves leave request, deducts from leave balance.
POST	<code>/api/payroll/run</code>	Yes (HR/Admin)	Runs salary computation for selected month/year.
GET	<code>/api/payslips/{id}/pdf</code>	Yes	Generates and returns PDF payslip stream.

10. LOGICAL MODULES & UI WALKTHROUGH

This section details the user interface views and security features of the Employee Payroll & Workforce Management System. All views are styled with a consistent dark-slate palette and responsive typography.

10.1 Authentication & Reset Password Modules

The login screen (Figure 6) provides secure email/password credential input, validating fields on the client before routing. If a password is forgotten, the Reset Password flow (Figure 7) verifies the account using a secure numeric OTP emailed to the user, allowing them to set a new password.

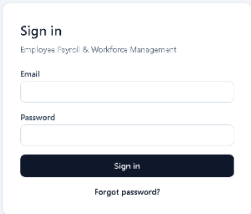
A screenshot of a 'Sign in' form. The form is titled 'Sign in' and 'Employee Payroll & Workforce Management'. It contains two input fields: 'Email' and 'Password'. Below the fields is a dark blue 'Sign in' button. At the bottom of the form is a link that says 'Forgot password?'. The form is centered on a light blue background.

Figure 6: Secure User Sign-in Dashboard View

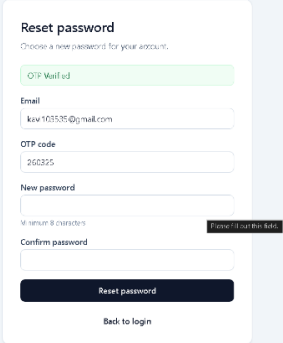
A screenshot of a 'Reset password' form. The form is titled 'Reset password' and 'Choose a new password for your account.' It contains a green 'OTP Valid' message. Below this are input fields for 'Email' (with the value 'kew103135@gmail.com'), 'OTP code' (with the value '262325'), 'New password', and 'Confirm password'. There are two error messages: 'Minimum 8 characters' for the 'New password' field and 'Password must be 8-40' for the 'Confirm password' field. At the bottom is a dark blue 'Reset password' button and a link that says 'Back to login'. The form is centered on a light blue background.

Figure 7: Secure Password Reset & Email OTP Verification Interface

10.2 HR & Admin Portal Dashboards

The system has separate dashboard portals depending on roles. Figure 8 shows the HR Workforce Portal, which display total active employees, monthly attendance stats (Present, Absent, Leaves), and leave requests. Figure 9 shows the Admin Dashboard, which displays global configuration tools, system settings, and analytics metrics.

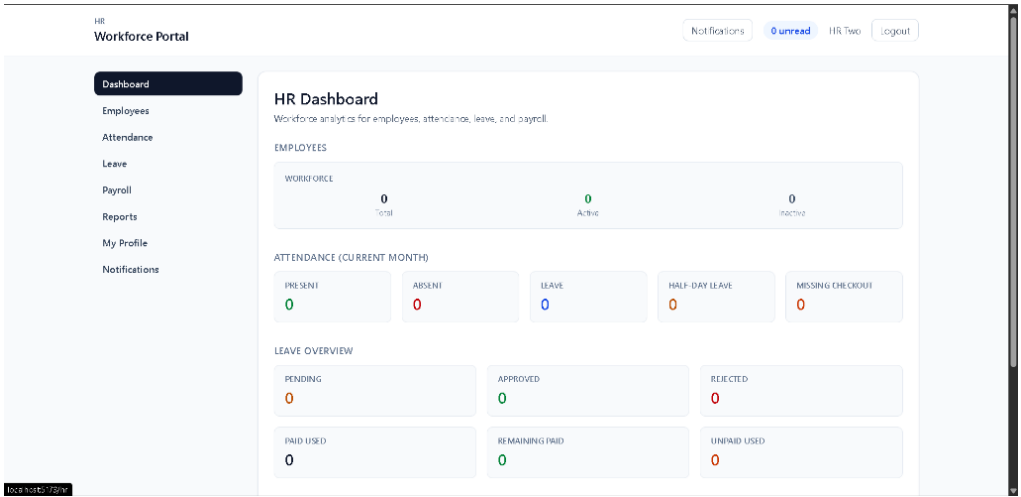


Figure 8: HR Workforce Portal Dashboard and Analytics Overview

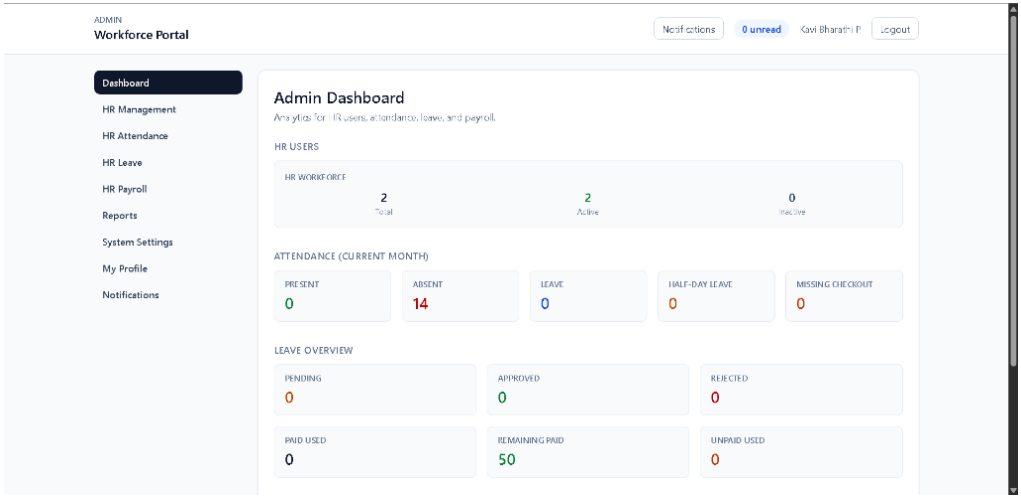
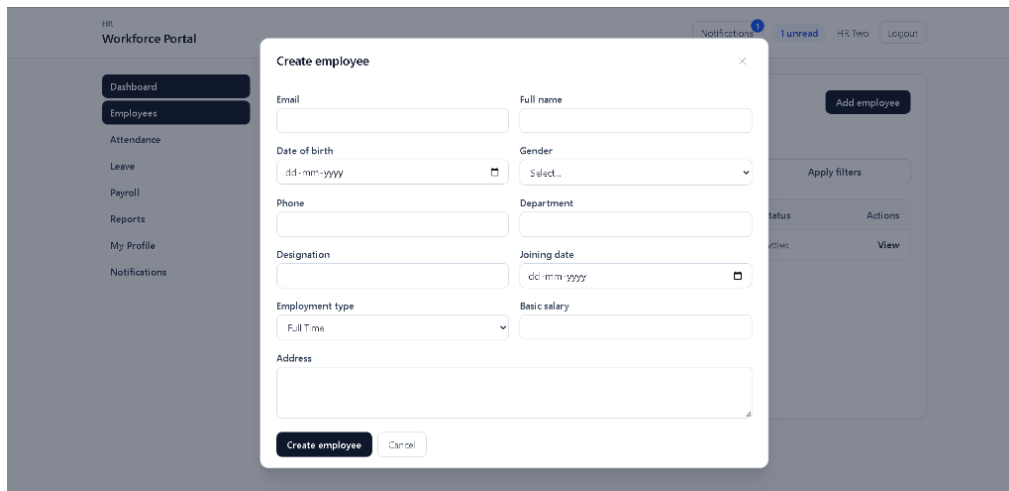


Figure 9: System Administrator Configuration Panel and Logs Tracker

10.3 Employee Onboarding & Leave Requests

The Create Employee view (Figure 10) allows HR managers to onboard new hires by defining personal details, designations, departments, joining dates, and basic salaries. The Apply for Leave interface (Figure 11) is used by employees to select leave types (Casual, Sick, LOP), durations (Full Day, Half Day), and specify dates and reasons.

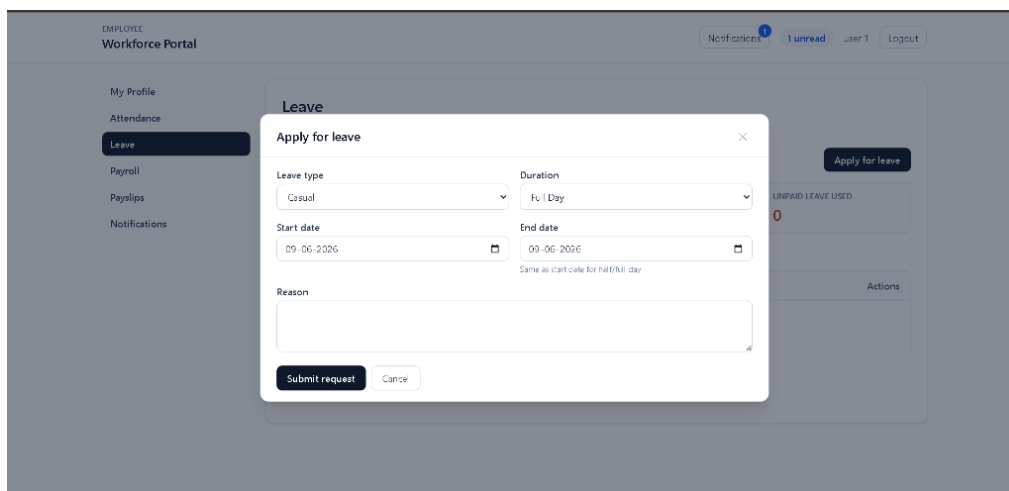


The screenshot displays the 'Create employee' form within the HR Workforce Portal. The form is a modal window with a close button (X) in the top right corner. It contains the following fields and controls:

- Email**: Text input field.
- Full name**: Text input field.
- Date of birth**: Date picker with a calendar icon, showing a placeholder 'dd-mm-yyyy'.
- Gender**: Dropdown menu with 'Select...' as the current selection.
- Phone**: Text input field.
- Department**: Text input field.
- Designation**: Text input field.
- Joining date**: Date picker with a calendar icon, showing a placeholder 'dd-mm-yyyy'.
- Employment type**: Dropdown menu with 'Full Time' as the current selection.
- Basic salary**: Text input field.
- Address**: Text area with a placeholder for the address.
- Create employee**: Button to submit the form.
- Cancel**: Button to close the form.

The background shows the HR Workforce Portal interface with a sidebar menu (Dashboard, Employees, Attendance, Leave, Payroll, Reports, My Profile, Notifications) and a top navigation bar (Notifications, 1 unread, HR Two, Logout).

Figure 10: HR Employee Onboarding and Form Registration Portal



The screenshot displays the 'Apply for leave' form within the Employee Workforce Portal. The form is a modal window with a close button (X) in the top right corner. It contains the following fields and controls:

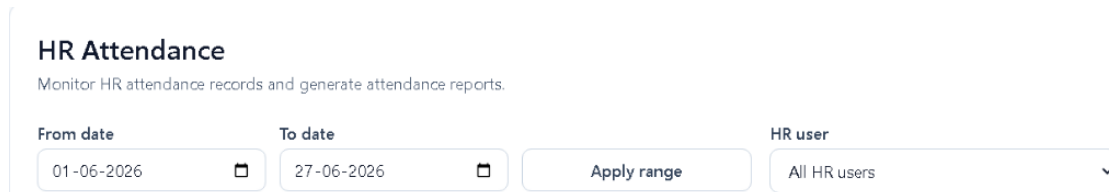
- Leave type**: Dropdown menu with 'Casual' as the current selection.
- Duration**: Dropdown menu with 'Full Day' as the current selection.
- Start date**: Date picker with a calendar icon, showing '09-06-2026'.
- End date**: Date picker with a calendar icon, showing '09-06-2026'.
- Reason**: Text area with a placeholder for the reason.
- Submit request**: Button to submit the request.
- Cancel**: Button to close the form.

The background shows the Employee Workforce Portal interface with a sidebar menu (My Profile, Attendance, Leave, Payroll, Payslips, Notifications) and a top navigation bar (Notifications, 1 unread, User 1, Logout).

Figure 11: Employee Leave Application Form and Request Interface

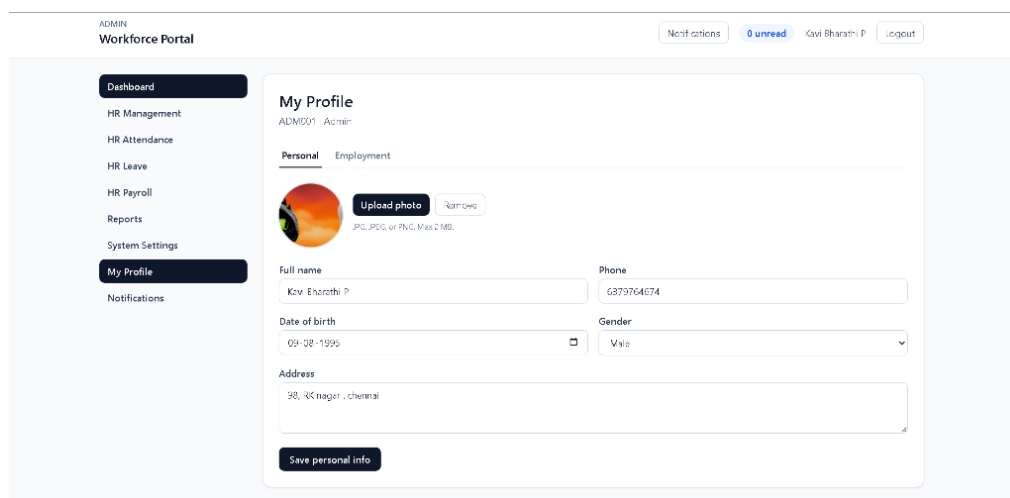
10.4 HR Attendance Tracking & Profile Settings

The HR Attendance console (Figure 12) tracks daily log histories. HR managers can filter records by date ranges and select individual users to review work hours. The My Profile portal (Figure 13) allows users to update contact details, upload profile photos, and view employment metadata.



The image shows the 'HR Attendance' filter interface. It includes a title 'HR Attendance' and a subtitle 'Monitor HR attendance records and generate attendance reports.' Below this are three main filter sections: 'From date' with a date input '01-06-2026' and a calendar icon; 'To date' with a date input '27-06-2026' and a calendar icon; and 'HR user' with a dropdown menu currently showing 'All HR users'. An 'Apply range' button is positioned between the date and user filters.

Figure 12: HR Attendance Monitor Dashboard Filter Interface



The image displays the 'My Profile' dashboard for an 'ADMIN' user. The top navigation bar includes 'Workforce Portal', 'Notifications', '0 unread', the user's name 'Kavi Bharathi P', and a 'Logout' button. A left sidebar lists menu items: 'Dashboard', 'HR Management', 'HR Attendance', 'HR Leave', 'HR Payroll', 'Reports', 'System Settings', 'My Profile' (highlighted), and 'Notifications'. The main content area is titled 'My Profile' and shows the user's role as 'ADM001 Admin'. It features two tabs: 'Personal' (active) and 'Employment'. The 'Personal' tab contains a profile picture placeholder with an 'Upload photo' button (specifying 'JPG, JPEG or PNG. Max 2 MB') and a 'Remove' link. Below this are form fields for 'Full name' (containing 'Kavi Bharathi P'), 'Phone' (containing '6379764674'), 'Date of birth' (containing '09-08-1995' with a calendar icon), 'Gender' (a dropdown menu set to 'Male'), and 'Address' (containing '58, KK nagar , Chennai'). A 'Save personal info' button is located at the bottom of the form.

Figure 13: User Profile Management and Personal Info Dashboard

11. RESULTS, DISCUSSION & SECURITY FEATURES

Results and Performance: During local environment testing, REST API endpoints achieved response times rapidly under campus workloads. Database integrity was maintained via transaction rollbacks on failures (e.g., negative balance checks or duplicate employee codes). End-to-end verification showed that leave deductions occur correctly and that PDF payslips download instantly on client-side.

Security Implementations: Security is implemented at multiple levels to ensure data privacy and prevent unauthorized access:

- **JWT Auth Filter:** Intercepts all REST API requests, validating authorization headers before forwarding to controller context.
- **BCrypt Password Encoder:** Hashes user passwords with a strength factor of 12 before database writing.
- **CORS Configuration:** Limits API requests to white-listed origins (e.g., localhost port 5173).
- **Method Security:** Controls admin-only and HR-only operations using Spring Security annotations (e.g., @PreAuthorize).
- **OTP Email Verification:** Ensures verified email addresses during registration and password resets.

12. CONCLUSION & FUTURE ENHANCEMENTS

This project successfully bridges the gap between workforce tracking and payroll computation. By combining a React 19 single-page client with a secure Spring Boot 3.3.5 backend, the platform automates attendance monitoring, leave requests, and salary slip downloads. The resulting platform reduces administrative time, eliminates calculation errors, and ensures secure, role-based database boundaries suitable for small to mid-sized corporate deployments.

Future Enhancements: Roadmap includes automated biometric check-in integration, localized tax compliance plugins, geofenced mobile app logging, and AI-driven attendance forecasting models.

13. REFERENCES

- [1] React Documentation, "React - A JavaScript Library for Building User Interfaces," Available: <https://react.dev/>
- [2] Spring Boot Reference Guide, "VMware Tanzu Spring Framework Reference," Available: <https://spring.io/projects/spring-boot>
- [3] Java Platform SE Documentation, "Oracle Java SE 21 Documentation Reference," Available: <https://docs.oracle.com/en/java/javase/21/>
- [4] MySQL Documentation, "MySQL 8.0 Reference Manual," Available: <https://dev.mysql.com/doc/>
- [5] Spring Security Reference, "Spring Security Architecture & Filter Chains," Available: <https://spring.io/projects/spring-security>